EFTHIMIOS VLAHOS

Senior Machine Learning Engineer | Applied AI / Forward-Deployed Engineer | AI Architect | LLM Agents, RAG, Evals
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SUMMARY

Senior Machine Learning Engineer with 6+ years building production AI systems and the last three shipping **LLM agents, retrieval and evaluation pipelines** in regulated industries (insurance, energy, manufacturing). Architect and hands-on builder of agentic platforms end to end: discovery with business stakeholders, system design, **multi-agent orchestration**, evals and guardrails, deployment and operations. Set the LLM evaluation, model-routing and cost standards that other teams adopt; co-founder of an AI platform company with paying customers. The part of the work I enjoy most is between the demo and production: sitting with the people who own the problem until it is precisely defined, then building the system and staying accountable for it once it ships. Targeting Forward-Deployed / Applied AI Engineer, AI Architect and AI engineering leadership roles.

EXPERIENCE

Senior Machine Learning Engineer — Alliant Insurance Services, Manhattan, NY (Hybrid) Feb 2026 – Present

- Architected and deployed **DocEngine**, an **agentic AI system integrated with Salesforce** that automates end-to-end insurance policy generation and processing for **10,000+ policy documents per month**, cutting turnaround from days to under an hour and saving the organization millions of dollars in operational costs.
- Designed **multi-step LLM agents** (planning, tool calling, self-validation against JSON schemas) shipped as production microservices with **human-in-the-loop approval gates** and full audit trails; agents resolve **80%** of requests without human escalation.
- Lead **LLM evaluation, model routing and cost optimization** across frontier APIs and fine-tuned open-weight models on a **500+ case golden eval set**; routing and prompt caching cut inference spend **40%**, and the eval framework and guardrails are the standard other teams adopt.
- Built the **AgentOps foundations**: distributed tracing, observability dashboards, **regression evals in CI** and cost ceilings for agents in production, halving agent failure rates and bringing **p95 latency under 10 seconds**.
- Built a **document-intelligence pipeline** over proprietary policy and claims documents (extraction, grounded retrieval, source citations) with **96% field-level extraction accuracy** against human review and every answer traceable to its source page.
- Partner with underwriting and operations stakeholders to translate business requirements into ML solutions; run **design reviews** and **mentor four engineers** on agent design, evals and safe deployment in a regulated industry.

Machine Learning Engineer — Con Edison, Manhattan, NY (Hybrid) Jan 2025 – Jan 2026

- Architected a **LangGraph text-to-SQL agent** with semantic schema routing (query embeddings select tables), SQL generation, execution and automated report generation; adopted by **100+ analysts**, cutting ad-hoc reporting time **40%** for non-technical stakeholders.
- Engineered the **propensity-to-enroll pipeline on Databricks** (Delta Live Tables ingestion, PySpark and Feature Store features, MLflow experiments with a champion XGBoost model, Unity Catalog registry, batch inference via Databricks Jobs); **85%** precision, driving **$2M+** revenue in Energy Efficiency campaigns.
- Built a hybrid **entity-resolution pipeline** for customer-name variations: fine-tuned Sentence Transformers embeddings, FAISS approximate-nearest-neighbor clustering, and LLM few-shot classification as the contextual judge for final matching, lifting match F1 **35%** over rules-based matching across **20+ source systems**.
- Built the **MLOps framework on Azure** (MLflow, Docker, Terraform, Kubernetes with Kubeflow Pipelines and KServe, Jenkins, Prometheus/Grafana) that standardized training, registry, deployment and observability, reducing model onboarding time **50%**.

Data Scientist (Contract) — Cornerstone Building Brands, East Meadow, NY (Remote) May 2024 – Mar 2025

- Designed and productionized an end-to-end **recommendation system on Databricks** (scikit-learn, feature store, MLflow tracking and registry) across **10,000+ SKUs** for dealer and contractor customers, generating **$3.5M** incremental revenue.
- Deployed a production **Flask application** (Docker, Kubernetes) delivering real-time quote-to-close predictions from tuned logistic regression and random forest models, increasing conversion rates **20%**.
- Built **ETL/ELT pipelines** in Databricks (PySpark, Delta Lake, SQL) from Azure Data Lake with Airflow orchestration and GitHub Actions CI/CD, reducing runtime **35%** and excess inventory costs **15%**; Power BI dashboards used by **50+** executives and plant managers.

Data Analyst (Contract) — Microsoft, East Meadow, NY (Remote) Oct 2020 – May 2024

- Analyzed enterprise-scale datasets (Python, SQL, R) supporting an **LLM-based chatbot**; anomaly detection and data cleansing cut hallucination rates **30%** and raised user satisfaction **25%**.
- Fine-tuned a **Phi small language model** (TensorFlow, PyTorch) and built knowledge-grounded chatbot prototypes with a **5,000+ conversation** evaluation set, improving contextual accuracy and cutting manual escalations **20%** for key clients.

ENTREPRENEURIAL EXPERIENCE

Co-Founder & AI Platform Lead — Pyxeon (side venture), Long Island, NY · pyxeon.com Mar 2026 – Present

- Architected the platform end to end: **8 live services** behind a Caddy reverse proxy with automatic HTTPS (marketing site, admin dashboard, customer portal, API, four customer apps), **multi-tenant isolation** with per-customer Supabase databases and encrypted GitHub tokens, Clerk auth, and Stripe subscription billing with per-tier request quotas.
- Built the **autonomous coding pipeline**: Claude Sonnet 5 analyzes each request from a tier-prioritized queue and returns a structured plan with effort estimate within minutes; on customer approval an agent clones the repo, implements, runs tests, commits and opens a PR; an **independent Opus 5 verifier** reviews the diff against the approved plan; every change ships with an evidence bundle (plain-English summary, before/after screenshots, checks) and **one-click rollback**.
- Designed the **Docker agent-scaling architecture** (Redis job queue, dispatcher launching resource-limited containers with HMAC-signed callbacks and a watchdog, concurrent runs, Kubernetes job manifests) and a **self-healing operations loop**: uptime checks every 5 minutes open a fix request automatically after 15 minutes of downtime and route it through the same pipeline.
- Own product, pricing and go-to-market: tiered subscriptions with monthly request quotas, an **AI consultation funnel** that qualifies business owners on the marketing site, and an **App Factory** that turns a business intake into a branded application; **two paying customers**, four customer-facing apps in production, median time from approved plan to deployed change **under 24 hours**, agent cost per shipped change **under $10**, **99.9%** monitored uptime; **1,500+ backend and 47 frontend tests**, ~600 commits since March 2026.

SELECTED PROJECTS

- **Trader Hub** — AI options and equities trading platform on Alpaca: **LangGraph strategy council** of six specialist agents, autonomous scanner with budget and cooldown guardrails, **graduation gate** (Sharpe at least 1, max drawdown at most 15%) before live trading, backtests with lookahead protection; **323 tests**. FastAPI, Next.js.
- **Blueprint Takeoff** — **vision-LLM application** that reads an electrical drawing's legend, counts every symbol across multi-page PDFs and produces a material takeoff in under a minute; Docker.
- **DeskSergeant** (independent consulting) — **multi-tenant fraud and AI-security platform** (FastAPI, Next.js 16, Neon Postgres, iOS); **1,287 backend tests**; 182 architecture decision records.

TECHNICAL SKILLS

Agentic AI: multi-agent orchestration (supervisor, hierarchical, planner / implementer / verifier), LangGraph (state, checkpointing, human-in-the-loop interrupts), Claude Agent SDK, OpenAI Agents SDK, tool and function calling, structured outputs, Model Context Protocol (MCP) servers and clients, A2A, agent memory and context engineering, prompt caching, sandboxed code execution, browser and computer-use agents, durable workflows, streaming

Agent Evaluation & Safety: trajectory and tool-call evals, task-completion metrics, golden datasets, LLM-as-judge with judge calibration, regression suites in CI, online evals and A/B tests, guardrails (prompt-injection defense, PII redaction, policy filters, output validation), red-teaming, cost ceilings and model routing

Retrieval & Knowledge: RAG and agentic RAG (chunking, hybrid BM25 + dense retrieval, cross-encoder reranking, query rewriting, grounding and citations), GraphRAG, Sentence Transformers, FAISS, Chroma, pgvector, Qdrant, semantic caching

Models, Fine-Tuning & Serving: Claude (Anthropic API), OpenAI, Phi, Llama, Mistral, Qwen; post-training: SFT, LoRA/QLoRA (PEFT), DPO/ORPO preference tuning, GRPO/RLVR, distillation, 4-bit quantization; Hugging Face Transformers, TRL, Unsloth, Axolotl; LiteLLM, OpenRouter, vLLM, KServe, batch inference; PyTorch, TensorFlow/Keras, scikit-learn, XGBoost

MLOps & AgentOps: MLflow (tracking, registry), Databricks (Delta Live Tables, Feature Store, Unity Catalog, Jobs), Kubeflow Pipelines, Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, Prometheus/Grafana, LLM tracing (LangSmith, Langfuse, Arize Phoenix, OpenTelemetry), Redis job queues

Cloud & Data: Azure (Machine Learning, Data Lake, AKS), AWS (SageMaker, S3, Glue, EKS), GCP (Vertex AI, BigQuery), Spark/PySpark, Delta Lake, Airflow, SQL, PostgreSQL (Supabase, Neon), Redis, Caddy/NGINX

Engineering: Python, TypeScript, SQL, Bash, R; FastAPI (async), Flask, React/Next.js, SQLAlchemy, REST APIs, JWT and Clerk auth, Stripe, HMAC-signed webhooks, multi-tenant isolation; pytest, test-first development, architecture decision records, CI/CD, AI-assisted engineering (Claude Code)

EDUCATION

Stony Brook University — Master of Science, Applied Mathematics & Statistics May 2023

CUNY Hunter College — Bachelor of Science, Mathematics & Physics; Minor in Computer Science May 2020

CERTIFICATIONS

Databricks Certified Data Engineer Associate (May 2025) | **AWS Certified Cloud Practitioner** (Apr 2024) | **LLMOps** and **MLOps**, Duke University / Coursera (Jun 2024, May 2024)